Assignment 4 – Ethical AI Questions

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Introduction

AI now shapes decisions that used to belong to people — hiring, driving, policing, and how a network allocates its own resources. This assignment answers thirty questions on the ethics of those systems: the core concepts and international frameworks, applied scenarios, and reflective questions about where AI ethics is heading and my role in it.

Section A: Conceptual Understanding

Question 1 — What is Ethical AI, and why is it essential in modern technological systems?

Ethical AI is the practice of designing and operating AI systems so they respect human rights, treat people fairly, protect privacy, and remain under meaningful human control. It is essential because AI makes high-stakes decisions at machine scale: a biased system discriminates against every affected person, consistently and invisibly, and only trustworthy systems earn public and regulatory acceptance.

Question 2 — List and explain any four major ethical principles that guide AI development.

Fairness — no systematically worse outcomes for people based on characteristics like gender or race. **Transparency** — those affected know AI is used and understand how it works. **Accountability** — identifiable people remain answerable for outcomes, with routes for appeal. **Privacy** — only necessary data is collected, protected, and used for its stated purpose (OECD, 2019; UNESCO, 2021).

Question 3 — What is the main objective of the IEEE Ethically Aligned Design framework?

To ensure autonomous and intelligent systems prioritize human well-being — not just performance — and that ethics is built in from the start of design, when ethical choices are still cheap to make, rather than patched on after deployment (IEEE, 2019).

Question 4 — Define the term algorithmic bias and provide one real-world example.

Algorithmic bias is a systematic tendency of an AI system to produce unfair outcomes for particular groups, usually inherited from unrepresentative or historically skewed training data. Example: Amazon's experimental recruiting tool, trained on a decade of male-dominated hiring data, penalized resumes containing the word "women's" and was scrapped (Dastin, 2018).

Question 5 — Explain the difference between transparency and explainability in AI.

Transparency is openness about the system itself — that AI is used, what data trained it, its purpose and limitations. Explainability is decision-level: giving a person an understandable reason for a specific output, such as why a loan was denied. A documented black box can be transparent but not explainable; meaningful recourse requires both.

Question 6 — What role does accountability play in the ethical use of AI systems?

It keeps a named person or organization answerable for the system's outcomes, enabling audits, appeals, and redress — so "the algorithm decided" never becomes an excuse. It also improves behavior upstream: teams that will answer for failures test more carefully before deploying.

Question 7 — Identify three international organizations involved in developing AI ethics guidelines.

UNESCO (Recommendation on the Ethics of AI), the OECD (AI Principles), and the IEEE (Ethically Aligned Design). Others include the European Union, ISO/IEC, and the ITU.

Question 8 — What does UNESCO's Recommendation on the Ethics of Artificial Intelligence emphasize?

Adopted by 193 member states in 2021 as the first global AI-ethics standard, it grounds AI in human rights and dignity and emphasizes proportionality and doing no harm, fairness, privacy, human oversight, transparency, accountability, and environmental sustainability. It opposes uses incompatible with human rights, such as social scoring and mass surveillance (UNESCO, 2021).

Question 9 — What are OECD's five key principles for trustworthy AI?

(1) Inclusive growth, sustainable development and well-being; (2) respect for human rights, democratic values and fairness; (3) transparency and explainability; (4) robustness, security and safety across the lifecycle; (5) accountability of those deploying AI (OECD, 2019).

Question 10 — How does the EU AI Act classify and regulate AI systems based on risk?

Four tiers: **unacceptable risk** is banned (social scoring, harmful manipulation, certain biometric surveillance); **high risk** (hiring, credit, law enforcement, critical infrastructure) requires risk management, quality data, logging, human oversight, and conformity assessment; **limited risk** carries transparency duties such as disclosing chatbots and labeling AI content; **minimal risk** has no new obligations. General-purpose AI models have their own requirements (European Union, 2024).

Section B: Applied and Scenario-Based

Question 11 — An AI recruitment tool consistently prefers male candidates. Which ethical principles are violated?

Primarily fairness and non-discrimination; also transparency, since rejected candidates never learn an algorithm screened them or why, and accountability if the tool stays in use unaudited. The Amazon case shows the cause is usually biased training history rather than intent — which is exactly why proactive bias testing is an obligation (Dastin, 2018).

Question 12 — A self-driving car must choose between hitting one pedestrian or five. Which ethical dilemma does this represent?

The trolley problem: utilitarian reasoning (minimize total harm) against deontological reasoning (there is a moral difference between allowing harm and actively choosing a victim). The Moral Machine experiment showed these preferences vary across cultures, so whatever gets coded is a value judgment that demands transparency and accountability (Awad et al., 2018).

Question 13 — In predictive policing, how can bias be introduced into the algorithm's training data?

Crime records measure enforcement, not crime: over-patrolled neighborhoods generate more recorded incidents, the model flags them as high-risk, more patrols follow, and the feedback loop reinforces itself. Label bias (arrests are not convictions) and proxy variables such as zip code add further distortion.

Question 14 — How might a "human-in-the-loop" system reduce ethical risks in automated decision-making?

A human reviews consequential outputs before they take effect — catching errors and context the model cannot see and preserving a clear point of accountability. The EU AI Act mandates such oversight for high-risk systems; it only works if the reviewer has genuine authority rather than rubber-stamping (European Union, 2024).

Question 15 — Describe how data privacy laws such as GDPR affect the design of AI systems.

GDPR requires a lawful basis and stated purpose, data minimization, and privacy by design, and grants rights of access, correction, and erasure that systems must be built to honor. Most importantly for AI, Article 22 restricts solely automated decisions with significant effects and requires meaningful information about the logic — pushing designs toward human review and explainability (European Union, 2016).

Question 16 — A generative AI creates a fake video of a politician. What ethical issues arise here?

Deception and manipulation of voters, use of a person's likeness without consent, potential defamation, and erosion of public trust — including the "liar's dividend," where genuine recordings become deniable as fakes. This is why the EU AI Act requires AI-generated content to be labeled (European Union, 2024).

Question 17 — What are the ethical implications of AI-powered surveillance systems in public spaces?

Everyone becomes tracked by default, eroding privacy and chilling expression and assembly, while facial recognition's higher error rates for darker-skinned women distribute misidentification unequally (Buolamwini & Gebru, 2018). Function creep and concentration of power compound this — which is why UNESCO opposes mass surveillance and the EU tightly restricts real-time biometric identification (European Union, 2024; UNESCO, 2021).

Question 18 — How can organizations ensure transparency in AI-driven financial decisions?

Disclose that AI is involved; give specific reasons with every adverse decision; prefer interpretable models or pair complex ones with reliable explanations; document and independently audit models, including performance across demographic groups; and provide appeal to a human with authority to overturn. Credit scoring is high-risk under the EU AI Act, so much of this is becoming law (European Union, 2024).

Question 19 — Discuss the ethical concerns surrounding AI in healthcare diagnostics.

Models trained on unrepresentative populations can be systematically less accurate for some patients; liability for errors must be clearly assigned among clinician, hospital, and vendor; clinicians should not act on predictions they cannot interrogate; health data demands the strongest privacy protection; and automation bias can erode human judgment. The consensus: clinically validated decision support under human oversight, not autonomous diagnosis.

Question 20 — What measures can be implemented to make AI datasets more representative and fair?

Audit dataset composition and test performance disaggregated by group, since averages hide unequal error rates (Buolamwini & Gebru, 2018); deliberately source and rebalance under-represented groups; document each dataset's contents, collection, and intended use; use diverse annotators; and keep monitoring after deployment. Fairness is a maintenance discipline, not a one-time preprocessing step.

Section C: Reflective and Future-Oriented

Question 21 — Do you think AI can ever be completely unbiased? Justify your answer.

No. Models learn from data produced by an imperfect world, and "unbiased" itself requires normative choices — reasonable mathematical definitions of fairness can be mutually incompatible, so someone must decide which one matters for a given system. The realistic goal is managed bias: measured, disclosed, minimized, and monitored.

Question 22 — How can ethical frameworks ensure responsible innovation in AI?

They translate values into design requirements, audits, and shared expectations, moving ethics upstream to design time when it is cheap, and set a common floor so no competitor gains by cutting corners. They work when paired with enforcement — the EU AI Act attaches penalties to what IEEE and the OECD express as principles (European Union, 2024; IEEE, 2019; OECD, 2019).

Question 23 — Should governments enforce stricter AI regulations, or should industries self-regulate? Discuss.

A hybrid, tiered by risk. Self-regulation alone fails under competitive pressure — someone always defects — while pure government control lags the technology. Governments should set a binding floor for high-stakes uses (bans, mandatory oversight, real penalties, as in the EU AI Act), while industry builds the fast-moving technical standards that regulators then reference.

Question 24 — What ethical challenges might Quantum AI pose in the future?

A capable quantum computer breaks today's public-key encryption — and adversaries already harvest encrypted data to decrypt later — so quantum-safe migration is an ethical duty to users, not just an upgrade. Quantum-accelerated AI would also concentrate capability in a few states and firms and be even harder to audit; for once, governance should arrive before the capability.

Question 25 — How might Generative AI reshape intellectual property rights?

On the input side, training on copyrighted works without consent or compensation is being tested in court, with licensing markets emerging in anticipation. On the output side, human-authorship requirements leave purely AI-generated work unprotected, making "how much human contribution is enough" the new boundary — alongside style imitation of living artists and a push toward provenance watermarking and disclosure.

Question 26 — What are the moral responsibilities of AI developers toward end-users?

Anticipate harm and misuse before release; test for bias and safety and state limitations honestly; protect user data as a duty; disclose when users are dealing with AI and provide recourse when it is wrong; and refuse applications that cannot be built responsibly. Software affecting millions deserves the same duty of care older engineering professions accepted long ago.

Question 27 — How can AI sustainability contribute to global ethical goals?

In both directions. AI's own energy and water footprint makes efficiency a moral choice — my edge-computing lab's distilled model was ten times smaller than the baseline, a difference measured in megawatts at edge scale. And AI is itself a sustainability tool, from O-RAN energy-saving applications to grid optimization, aligning with UNESCO's and the OECD's sustainability principles (OECD, 2019; UNESCO, 2021).

Question 28 — What future jobs might emerge to address ethical issues in AI systems?

AI auditors certifying systems for bias and conformity; responsible-AI officers; AI Act compliance specialists (the analog of GDPR data-protection officers); red-team specialists paid to break models before adversaries do; model-documentation specialists; and AI incident responders applying the discipline my ITSM field uses for outages. Every technology that became critical infrastructure grew an assurance profession; AI is next.

Question 29 — In your opinion, what will be the most pressing AI ethics issue in 2035, and why?

Accountability for autonomous AI agents. The networks in this course already predict their own traffic and heal their own failures, and the same agentic pattern is spreading across the economy. By 2035 the central question will be who answers when chains of autonomous decisions, made at machine speed with no human in the loop, cause real harm — requiring decision logging, liability regimes for delegated authority, and limits on unsupervised automation.

Question 30 — Reflect on your role as a future professional — how would you ensure ethical practices in AI development?

My path runs through operating AI systems, and operations is where ethics lives or dies. I would hold systems to evidence, not demos — this course's target-leakage bug, where a perfect R^2 of 1.0 was actually a defect, taught me to treat too-good metrics as bug reports. I would apply ITSM discipline to models (change management, drift monitoring, honest incident review), insist on human oversight and appeal wherever a system touches people's opportunities, and treat documentation and reproducibility — as practiced across this portfolio — as everyday ethical acts.

Conclusion

The world's AI-ethics frameworks broadly agree on fairness, transparency, accountability, privacy, human oversight, and sustainability — the hard part is operational, as the scenarios in Section B show. The frontier is moving from auditing individual models to governing autonomous agents, and my takeaway from this course is that the intelligent network and the ethical network are the same engineering problem: both are only as trustworthy as the evaluation, oversight, and accountability built around them.

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